

SALTDRAW · REQUIREMENTS

SaltDrag requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdrag		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 status from SDG-CAL-001 against the constructable design (SDG-DDR-002); R5 not verified
0.2	2026-10-03	Amish Chadha	TRL 2; every target made measurable; R10 to R12 added for rope, holding and rope capacity (SDG-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

SaltDrag requirements

Eleven of the twelve requirements are met on paper by the constructable design; R5, the gathering rate against the dantala, cannot be shown until the dantala baseline is measured on a partner's pan. Status is from the sizing note SDG-CAL-001 (estimates). Physical verification is TRL 4 and TRL 5 work.

Table 1. Requirements and their status at TRL 3.

ID	Requirement	Target	Verification (TRL 4 or later)	TRL 3 status
R1	Work from the bund	A full harvest pass, including set-up and moving to the next strip, with no worker entering the pan	Observed field trial on a partner pan	Met by design: crews stand on dry ground behind the bunds; the return pull re-aligns the rake
R2	Manageable crank effort	120 N or less at each of two handles for the design rake load	Spring scale on the handle during trials	Met: 89 N estimated [B3]
R3	Pull capacity	At least 3 kN rope pull from two people at 120 N; every load-bearing part at least 2 to 1 on yield or rated strength at 6 kN	Proof-load test with CalRig	Met: 4.0 kN at the end of a 30 m pass, 3.7 kN on a full drum [B4], [B5]; lowest margin 2.1 [C2b]
R4	Protect the pan floor	Blade edge 5 to 25 mm above the floor; no steel in the pan; skid pressure 15 kPa or less; no visible cutting or cracking after 20 passes	Inspection of a test pan after trial passes	Met by design: 10 mm edge, 10.7 kPa [D1], [D2]; floor check at TRL 5
R5	Gathering rate	At least as many kilograms per worker-hour as the dantala method on the same pan	Side-by-side timed harvest trial	Not verified: no baseline yet; estimate 150 kg per worker-hour [D9]
R6	Portable	Heaviest single part 25 kg or less; every part of the set carried by up to four people	Weigh parts; timed move between pans	Met: heaviest 23.8 kg (a ballast bag filled on site), drum 21.3 kg [E1]
R7	Corrosion resistance	No loss of function after one season in brine and salt with a daily freshwater rinse	Season-long field trial with inspection log	Met by design: galvanised steel above the bund; FRP, UHMW-PE and 316 stainless in the pan
R8	Quick set-up	Capstans anchored and rake rigged in 20 minutes or less (two crews at once)	Timed trials	Met: 19 min estimated [D10], small margin
R9	Value engineering	Parts cost of the full set against the value-engineering target of USD 4,000	Costed bill of materials	USD 2,803, USD 1,197 under the target [E2], [E3]
R10	Rope safety	Rope minimum breaking strength, after splices, at least 5 times the rated 3 kN pull	Rope certificate; proof load	Met: 7.7 [C8]
R11	Holding	Pawl holds 6 kN; brake holds 3 kN with a hand force of 150 N or less; each ground anchor holds at least 1.5 times its share of the rated pull (proof-loaded on site before use)	Proof-load test with CalRig; anchor pull test on site	Met on paper: pawl boss margin 4.7 [C5b], brake 92 N [C7], anchor 2.4 times its 6 kN chain force [C10b] (soil to confirm)
R12	Pan coverage	Each drum stores at least 45 m of rope, for pans up to 30 m between bund crests	Wind on and measure	Met: 58 m in four layers, 38 mm flange freeboard [B8], [B9]

■ Assumptions

- The drag of a rake on a crystallised bed is about 0.9 kN per metre of blade with a full heap (SDG-CAL-001, Section A); this is the main uncertainty and is measured with the partner before the first build is loaded.
- Hand capstans are not treated as machines under Agariya registration conditions; written clarification is sought through the partner before any trial.

- Several families can share one set across a harvest season.
- Salt gathered into the windrow, about 3.3 m from the bund toe, is loaded with existing methods (out of scope).
- The pans have dry, firm ground behind the bunds on the two sides where the capstans stand.

Safety: R3, R10 and R11 are safety requirements. A load-bearing part that misses its margin, a rope without a certificate, or an anchor that fails its proof load on site stops the work until it is replaced (SDG-BLD-001, Safety stops).